How Does Female Leadership Relate to Wage–Productivity Gaps in Colombia?

Ingri Katherine Quevedo Rocha

George Washington University

March 26, 2026

Abstract

This study examines whether female leadership is associated with lower labor monopsony power in Colombian manufacturing firms. In labor markets with frictions such as search costs, limited mobility, and employer concentration, firms may exercise monopsony power and pay wages below workers' marginal productivity, generating wage-productivity gaps known as labor markdowns.

While these distortions arise from structural labor market frictions, firms differ substantially in how they exercise wage-setting power. Management decisions play an important role in shaping wage-setting practices and employment policies, suggesting that leadership characteristics may influence the extent to which firms exploit monopsony power. Prior research suggests that female leaders may adopt organizational practices that place greater weight on worker welfare, employment stability, and equitable compensation.

Using firm-level data from the Colombian Manufacturing Survey (1998-2024), I construct firm-level labor markdowns and relate them to the share of firm ownership held by women using regressions that control for firm and industry characteristics. Markdown estimates are derived from production-function methods that recover labor elasticities and compare the marginal revenue product of labor to wages.

The results show a negative association between female ownership shares and labor markdowns. These findings suggest that leadership composition may influence firms' wage-setting behavior and have implications for policies promoting greater female representation in firm leadership in emerging economies.

JEL Classification: D43, D63, J16,J42,J7, M12, M52

Keywords: female leadership, labor market power, wage–productivity gap, markdowns, monopsony, Colombia

1 Introduction

Standard microeconomic theory predicts that in competitive labor markets wages equal workers' marginal products. Under this benchmark, firms act as wage takers, workers can freely move across jobs, and compensation accurately reflects productivity. Departures from this framework arise when firms possess wage-setting power, as in monopsonistic labor markets where employers face upward-sloping labor supply curves (Robinson, 1969). A growing literature shows that labor markets are often characterized by frictions such as search costs (Card, 2022; Azar et al., 2019; Lentz and Mortensen, 2010; Moscarini and Postel-Vinay, 2010; Manning, 2021), limited job mobility (Caldwell and Danieli, 2024; Schubert et al., 2025; Starr et al., 2021; McAdams, 2019), and employer concentration (Azar et al., 2017; Benmelech et al., 2018; Rinz, 2022). These frictions grant firms wage-setting power, allowing them to pay wages below workers' marginal productivity and generating a wage-productivity gap commonly referred to as a labor markdown.

Evidence suggests that such distortions are particularly pronounced in developing and emerging economies, where labor market frictions tend to be stronger. In Latin America and the Caribbean, workers receive roughly half of the value they generate on average, compared to about 65 percent in the United States and 81 percent in advanced economies (Alviarez et al., 2025). For Colombia, Amodio and de Roux, 2024 estimate a labor supply elasticity of 2.5, implying that workers produce approximately 40 percent more than they earn.

While monopsony power arises from structural features of labor markets, firms differ substantially in how they exercise wage-setting power. Management decisions shape wage-setting practices, organizational culture, and employment policies, suggesting that firm leadership may influence the extent to which monopsony power is exercised. This paper examines whether leadership characteristics—specifically gender composition—are systematically associated with differences in labor market power.

In particular, I investigate whether firms with greater female ownership exhibit lower labor markdowns than firms primarily owned by men. Prior research suggests that female leaders may adopt management styles that place greater weight on worker welfare and employment stability (Matsa and Miller, 2013; Tunyi et al., 2023), interpret productivity signals differently (Flabbi et al., 2017; Chen et al., 2019), or influence corporate decision-making in ways that reduce wage distortions (Levi et al., 2014; Flabbi et al., 2019). Moreover, labor market power has been shown to affect women more strongly than men (Sharma, 2023), raising the possibility that female leadership may mitigate gender-specific wage distortions within firms. If female leaders place relatively greater weight on worker welfare, long-term employment relationships, or equitable compensation, firms they lead may exercise monop-

sony power less aggressively, resulting in lower labor markdowns.

To examine this relationship, I use firm-level data from the Colombian Manufacturing Survey (EAM, from its Spanish initials) collected by the National Department of Statistics of Colombia (Departamento Administrativo Nacional de Estadística (DANE), n.d.). I estimate labor markdowns following the production-function methodology developed by Yeh et al., 2022. Specifically, markdowns are calculated using elasticities derived from a Cobb-Douglas revenue production function, with the wage bill used as the measure of labor input. Following Ackerberg et al., 2015, estimates of the revenue function elasticity allow the marginal revenue product of labor to be computed and compared to the average wage paid by the firm, yielding a firm-level wage markdown. This is the same methodology used by the Inter-American Development Bank in its Development in the Americas (DIA) 2025 report and the CompeteLAC dataset for markdown estimation. I then estimate regressions relating firm-level labor markdowns to the share of firm ownership held by women, controlling for observable firm and industry characteristics.

Preliminary evidence using industry-level markdowns estimated by Inter-American Development Bank, 2025 suggests a negative relationship between female ownership and labor markdowns, although the estimates are imprecise. Ongoing work will extend the analysis to firm-level markdowns and examine whether the relationship varies with labor market concentration and productivity differences across firms.

2 Literature Review

Table 1: Summary of Related Literature

Paper	Topic	Data	Main Finding
De Loecker, Eeckhout, & Unger (2020)	Labor market concentration	U.S. manufacturing	Production approach for a robust estimation of elasticities. Findings suggest markups rise not just to cover higher fixed costs, but also due to firms' market power.
Yeh, Macaluso & Hershbein (2022)	Labor markdown estimation	U.S. manufacturing	Most manufacturing plants operate in a monopsonistic environment. Average markdown 1.53 workers earning only 65 cents on the marginal dollar generated.
Amodio & de Roux (2024)	Colombia Labor market power	Colombian manufacturing plants (EAM)	Estimate a firm-level labor supply elasticity of around 2.5, implying that workers produce about 40% more than their wage level
Inter-American Development Bank (2025)	Latin-America and the Caribbean Market Power	Firm-Data LAC (For Colombia EAM)	On average, workers in Latin America and the Caribbean take home in the form of wages 50 percent of what they contribute to their employers.
Matsa & Miller (2013)	Female leadership	Nordic companies (Norway's 2006 quota)	Female leaders tend to emphasize employment stability and worker retention.
Flabbi et al. (2019)	Female executives	Italy employer-employee panel data	Female leadership influences wage-setting and reduce gender gaps within firms.
Levi, Li & Zhang (2014)	Female executives	Aquisition bids by S&P 1500 companies	Female directors less overestimate merger gains and help create shareholder value

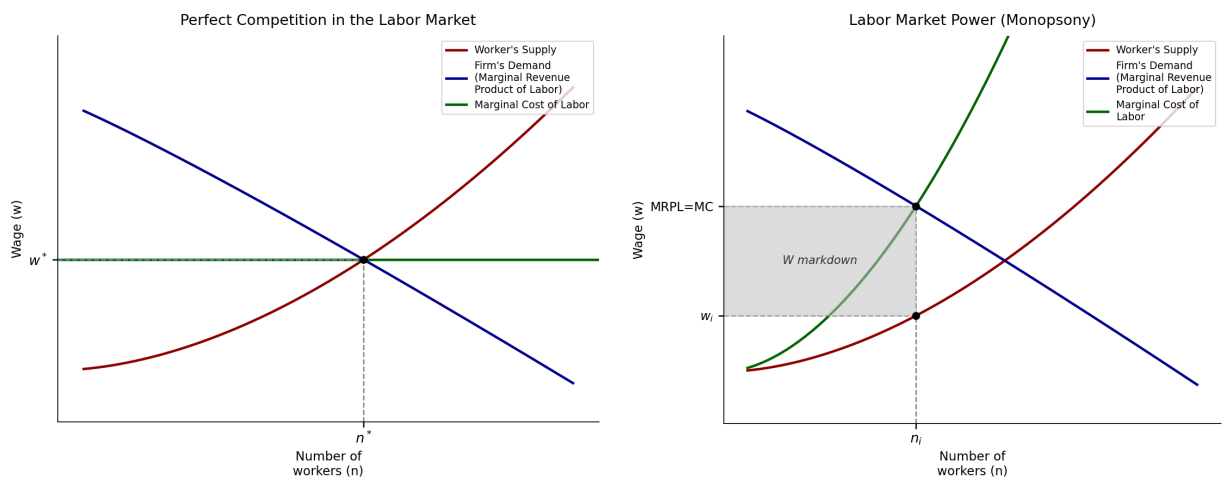
3 Theoretical Framework

To understand the wedge between wages and workers' marginal productivity, a useful starting point is the benchmark case of a competitive labor market. As shown in Panel (a) of Figure 1, under perfect competition the marginal cost of labor is constant and equal to the market wage, which each firm takes as given. Firms therefore hire labor up to the point where the marginal revenue product of labor (MRPL) equals the wage.

Under monopsony, by contrast, the firm faces an upward-sloping labor supply curve, as shown in Panel (b) of Figure 1. Hiring an additional worker requires raising the wage not only for the marginal worker but, when wages are set uniformly, for all inframarginal workers as well. As a result, the marginal cost of labor lies strictly above the labor supply curve. The profit-maximizing firm sets employment where MRPL = MCL, yielding both lower employment and a lower wage relative to the competitive benchmark. Crucially, at the optimum, workers are paid a wage w_i determined by the labor supply curve, yet their marginal contribution to firm revenue satisfies MRPL $> w_i$. The wedge between the marginal revenue product of labor and the wage—illustrated by the shaded region in Panel (b)—is the *markdown*, the central object of interest in this paper.

Figure 1: Labor Markets

(a) Perfect Competition (b) Monopsony



This paper follows Yeh et al., 2022 to estimate markdowns. Consider a firm's profit maximization problem:

$$\max_{L_{it}} \; R_{it}(L_{it}) - w_{it}(L_{it})L_{it}, \tag{1}$$

where $R_{it}(L_{it}) \equiv \text{rev}\Big(L_{it}; M^*_{-L}(L_{it})\Big)$ is firm i's revenue in year t, with all inputs evaluated at their optimum except labor L_{it}. Under monopsony, the firm faces an upward-sloping labor supply curve, so the relevant marginal cost of hiring is not the wage itself but the marginal expenditure on labor.

The firm's optimality condition is:

$$\frac{\partial R_{it}(L_{it})}{\partial L_{it}} = w_{it}(L^*_{it}) \left(1 + \frac{1}{\varepsilon^L_{it}}\right), \tag{2}$$

where $\varepsilon^L_{it} \equiv \frac{\partial w_{it}}{\partial L_{it}} \frac{L^*_{it}}{w_{it}(L^*_{it})}$ is the firm-level labor supply elasticity. The markdown ν_{it} is then defined as the ratio of the MRPL to the wage:

$$\nu_{it} \equiv \frac{\text{MRPL}_{it}}{w_{it}(L^*_{it})} = 1 + \frac{1}{\varepsilon^L_{it}}, \tag{3}$$

where $\text{MRPL}_{it} = \frac{\partial R_{it}(L_{it})}{\partial L_{it}}$. Under this definition, $\nu_{it} = 1$ under perfect competition (infinite labor supply elasticity) and $\nu_{it} > 1$ whenever labor supply is upward-sloping.

A complication noted by Yeh et al., 2022 is that a firm's perceived labor supply elasticity conflates labor market power (monopsony) with output market power (monopoly). To disentangle these, they adopt the production approach of De Loecker et al., 2020.

Consider a Cobb-Douglas production function with three inputs—labor L_{it}, capital K_{it}, and materials M_{it}—in an environment with both output and labor market power. Following De Loecker et al., 2020, variable inputs (L_{it}, M_{it}) adjust frictionlessly within a period, while capital is subject to adjustment frictions.

The cost minimization problem is:

$$\mathcal{L}_{it} = r_{it}K_{it} + w_{it}(L_{it})L_{it} + p^m_{it}M_{it} + F_{it} - \lambda_{it}\left[L^{\theta^L_{it}}_{it} K^{\theta^K_{it}}_{it} M^{\theta^M_{it}}_{it}\, \omega_{it} - \bar{Q}_{it}\right], \tag{4}$$

where λ_{it} is the Lagrange multiplier on the production constraint, interpreted as marginal cost. The firm's markup—the price-to-marginal-cost ratio—is therefore:

$$\mu_{it} = \frac{P_{it}}{\lambda_{it}}. \tag{5}$$

The first-order conditions for L_{it} and M_{it} are:

$$w_{it}(L^*_{it}) \left(1 + \frac{1}{\varepsilon^L_{it}}\right) = \theta^L_{it} \frac{P_{it}Q_{it}}{\mu_{it}L_{it}}, \tag{6}$$

$$p^m_{it}M_{it} = \theta^M_{it} \frac{P_{it}Q_{it}}{\mu_{it}}. \tag{7}$$

Defining the revenue shares of labor and materials as:

$$\alpha_{it}^{L} \equiv \frac{w_{it}(L_{it})L_{it}}{P_{it}Q_{it}}, \qquad \alpha_{it}^{M} \equiv \frac{p_{it}^{m}M_{it}}{P_{it}Q_{it}}, \tag{8}$$

and rearranging the first-order conditions yields closed-form expressions for the markup and markdown:

$$\mu_{it} = \frac{\theta_{it}^{M}}{\alpha_{it}^{M}}, \tag{9}$$

$$\nu_{it} = \frac{\theta_{it}^{L}}{\alpha_{it}^{L}} \cdot \frac{1}{\mu_{it}}. \tag{10}$$

4 Data

This paper uses firm-level data from the Annual Manufacturing Survey (Encuesta Anual Manufacturera, EAM) conducted by the National Administrative Department of Statistics (DANE) in Colombia. The dataset spans 1998-2019 and contains 175,914 observations, covering approximately 7,000 manufacturing firms per year. The EAM is a census of medium and large manufacturing establishments and provides detailed and standardized information on firms' production, input use, employment composition, and financial accounts.

The data are organized as an unbalanced panel, reflecting firm entry and exit over time. Following DANE's technical guidelines, the sample is restricted to firms with at least 10 employees, which ensures consistency in reporting standards and improves the reliability of firm-level measures. Observations with missing or implausible values (e.g., negative output or extreme input ratios) are excluded (0.2% of total observations). The data are inspected for outliers to ensure consistency with economic magnitudes.

The main outcome variable is the firm-level markdown, defined as the wedge between the marginal product of labor and the wage paid by the firm. Markdown estimates are obtained from a production function framework that combines estimated output elasticities with observed input shares; the construction is described in detail in the economic theory section.

The dataset provides the necessary information to construct standard production inputs using the methodology by De Loecker et al., 2020. Labor is measured as total employment, defined as the sum of permanent employees, firm owners, and temporary workers, both directly hired and outsourced. Capital is proxied by the book value of fixed assets, while the wage bill corresponds to total remuneration paid to permanent and temporary employees. Materials are measured as total expenditure on raw materials, intermediate inputs, and

packaging, and are treated as the flexible input in production which allows to control for unobservables (Levinsohn and Petrin, 2003). Gross output is proxied by firm-level total sales at factory's price without direct taxes as reported in the survey.

All monetary variables are expressed in nominal Colombian pesos. The empirical analysis relies primarily on ratios and input shares constructed from these variables, which mitigates concerns related to price-level variation. In addition, the dynamic structure of the production function estimation implies that rescaling variables through deflation would not materially affect the estimation of key parameters.

The main explanatory variable is the share of female ownership at the firm level. This measure is constructed using variables C4R1C1-C4R1C5, which report the number of firm owners (*owners, partners, and unpaid family members*) disaggregated by gender and worker type (white-collar and blue-collar). The female ownership share is defined as:

$$\text{Female Ownership Share}_{it} = \frac{\text{Number of female owners}_{it}}{\text{Total number of owners}_{it}}. \tag{11}$$

This variable ranges from 0 to 1 and captures the gender composition of firm ownership. Ownership is measured annually and may vary over time; firms with zero reported owners are excluded.

Finally, the dataset includes detailed identifiers for industry (sector) and geographic location (region), which are used to control for heterogeneity in production technologies, input markets, and regional economic conditions.

5 Empirical Framework

To examine the relationship between female ownership and labor market power, I estimate the following linear regression:

$$\ln \nu_{it} = \beta \, \text{Female}_{it} + \mathbf{X}_{it}'\boldsymbol{\gamma} + \alpha_t + \alpha_s + \alpha_r + \varepsilon_{it}, \tag{12}$$

where ν_{it} is the markdown estimated in Section the Theorical Framework Section for firm i in year t. The outcome is expressed in logs so that β has the interpretation of a semi-elasticity: the percentage difference in markdowns between female-owned and male-owned firms, conditional on controls.

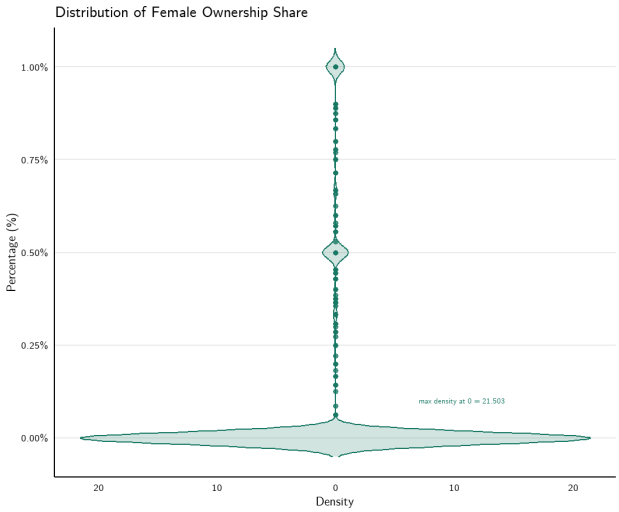
The main variable of interest, Female_{it}, measures the share of female ownership of the firm. The coefficient β therefore captures the conditional association between female ownership and the firm's degree of monopsony power.

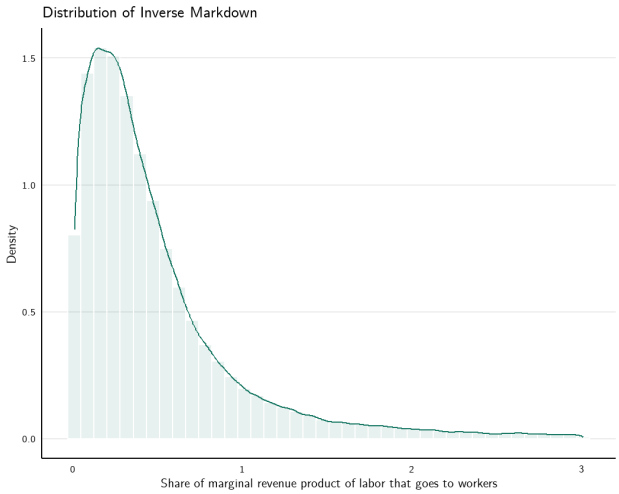
The vector $\mathbf{X}_{it}$ includes a set of firm-level controls. Firm age accounts for the possibility

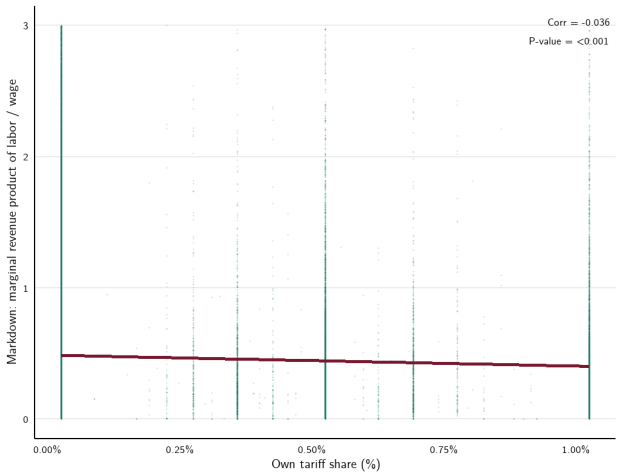
that older firms have had longer to build labor market power through reputation or search frictions. Firm size, measured by total revenue or total assets, controls for scale effects on both the production side and the firm's capacity to set wages. Finally, I include the share of female workers and the share of skilled workers in the firm's workforce to account for differences in employment composition that may independently affect markdowns.

The specification includes three sets of fixed effects. Year fixed effects α_t absorb common macroeconomic shocks and aggregate trends in markdowns over time. Sector fixed effects α_s account for permanent differences across industries in production technology, product market structure, and average labor market competitiveness. Region fixed effects α_r control for local labor market conditions, including differences in the outside options available to workers. Standard errors are clustered at the firm level to allow for arbitrary serial correlation within firms over time.

The coefficient β is identified from within-sector, within-region, within-year variation in female ownership across firms. The expected sign for the coefficient is $\beta < 0$ which would indicate that greater female ownership is associated with lower markdowns, consistent with female-owned firms exercising less monopsony power over their workers, conditional on observable firm characteristics.







References

Ackerberg, D. A., Caves, K., & Frazer, G. (2015). Identification Properties of Recent Production Function Estimators. *Econometrica*, *83*(6), 2411–2451. https://doi.org/10.3982/ECTA13408

Alviarez, V., Busso, M., Keefer, P., Santos, C., & Stucchi, R. (Eds.). (2025, December). *Markets for Development: Improving Lives through Competition*. Inter-American Development Bank. https://doi.org/10.18235/0013850

Amodio, F., & de Roux, N. (2024). Measuring Labor Market Power in Developing Countries: Evidence from Colombian Plants. *Journal of Labor Economics*, *42*(4), 949–977. https://doi.org/10.1086/725248

Azar, J., Huet-Vaughn, E., Marinescu, I., Taska, B., & Von Wachter, T. (2019, July). *Minimum Wage Employment Effects and Labor Market Concentration* (tech. rep.

No. w26101). National Bureau of Economic Research. Cambridge, MA. https://doi.org/10.3386/w26101

Azar, J., Marinescu, I., & Steinbaum, M. (2017, December). *Labor Market Concentration* (tech. rep. No. w24147). National Bureau of Economic Research. Cambridge, MA. https://doi.org/10.3386/w24147

Benmelech, E., Bergman, N., & Kim, H. (2018, February). *Strong Employers and Weak Employees: How Does Employer Concentration Affect Wages?* (Tech. rep. No. w24307). National Bureau of Economic Research. Cambridge, MA. https://doi.org/10.3386/w24307

Caldwell, S., & Danieli, O. (2024). Outside Options in the Labour Market. *Review of Economic Studies*, *91*(6), 3286–3315. https://doi.org/10.1093/restud/rdae006

Card, D. (2022). Who Set Your Wage? *American Economic Review*, *112*(4), 1075–1090. https://doi.org/10.1257/aer.112.4.1075

Chen, J., Leung, W. S., Song, W., & Goergen, M. (2019). Why female board representation matters: The role of female directors in reducing male CEO overconfidence. *Journal of Empirical Finance*, *53*, 70–90. https://doi.org/10.1016/j.jempfin.2019.06.002

De Loecker, J., Eeckhout, J., & Unger, G. (2020). The Rise of Market Power and the Macroeconomic Implications*. *The Quarterly Journal of Economics*, *135*(2), 561–644. https://doi.org/10.1093/qje/qjz041

Departamento Administrativo Nacional de Estadística (DANE). (n.d.). Encuesta Anual Manufacturera (EAM). https://www.dane.gov.co/index.php/estadisticas-por-tema/industria/encuesta-anual-manufacturera-enam

Flabbi, L., Macis, M., Moro, A., & Schivardi, F. (2019). Do Female Executives Make a Difference? The Impact of Female Leadership on Gender Gaps and Firm Performance. *The Economic Journal*, *129*(622), 2390–2423. https://doi.org/10.1093/ej/uez012

Flabbi, L., Piras, C., & Abrahams, S. (2017). Female corporate leadership in Latin America and the Caribbean region: Representation and firm-level outcomes. *International Journal of Manpower*, *38*(6), 790–818. https://doi.org/10.1108/IJM-10-2015-0180

Inter-American Development Bank. (2025). CompeteLAC. https://competelac.fislac.com/

Lentz, R., & Mortensen, D. T. (2010). Labor Market Models of Worker and Firm Heterogeneity. *Annual Review of Economics*, *2*(1), 577–602. https://doi.org/10.1146/annurev.economics.102308.124511

Levi, M., Li, K., & Zhang, F. (2014). Director gender and mergers and acquisitions. *Journal of Corporate Finance*, *28*, 185–200. https://doi.org/10.1016/j.jcorpfin.2013.11.005

Levinsohn, J., & Petrin, A. (2003). Estimating Production Functions Using Inputs to Control for Unobservables. *Review of Economic Studies*, *70*(2), 317–341. https://doi.org/10.1111/1467-937X.00246

Manning, A. (2021). The Elusive Employment Effect of the Minimum Wage. *Journal of Economic Perspectives*, *35*(1), 3–26. https://doi.org/10.1257/jep.35.1.3

Matsa, D. A., & Miller, A. R. (2013). A Female Style in Corporate Leadership? Evidence from Quotas. *American Economic Journal: Applied Economics*, *5*(3), 136–169. https://doi.org/10.1257/app.5.3.136

McAdams, J. M. (2019). Non-Compete Agreements: A Review of the Literature. *SSRN Electronic Journal*. https://doi.org/10.2139/ssrn.3513639

Moscarini, G., & Postel-Vinay, F. (2010). Unemployment and Small Cap Returns: The Nexus. *American Economic Review*, *100*(2), 333–337. https://doi.org/10.1257/aer.100.2.333

Rinz, K. (2022). Labor Market Concentration, Earnings, and Inequality. *Journal of Human Resources*, *57*(S), S251–S283. https://doi.org/10.3368/jhr.monopsony.0219-10025R1

Robinson, J. (1969). Monopsony. In *The Economics of Imperfect Competition* (pp. 218–228). Palgrave Macmillan UK. https://doi.org/10.1007/978-1-349-15320-6_19

Schubert, G., Stansbury, A., Taska, B., & Dey, M. (2025). Employer Concentration and Wages: Evidence from Large Firms' Hiring Shocks. https://doi.org/10.2139/ssrn.5374463

Sharma, G. (2023). *Monopsony and Gender* (Working Paper). Retrieved February 18, 2026, from https://economics.mit.edu/sites/default/files/inline-files/monopsony_gender_gsharma_16.pdf

Starr, E., Prescott, J. J., & Bishara, N. (2021). The Behavioral Effects of (Unenforceable) Contracts†. *The Journal of Law, Economics, and Organization*, *36*(3), 633–687. https://doi.org/10.1093/jleo/ewaa018

Tunyi, A. A., Areneke, G., Tob-Ogu, A., & Khalid, S. (2023). Doing more with more: Women on the board and firm employment. *Journal of Business Research*, *154*, 113385. https://doi.org/10.1016/j.jbusres.2022.113385

Yeh, C., Macaluso, C., & Hershbein, B. (2022). Monopsony in the US Labor Market. *American Economic Review*, *112*(7), 2099–2138. https://doi.org/10.1257/aer.20200025